Quiz Review: Proportions and Absolute Value

Name: _____ Date: _____

Part 1: Proportion Equations

Steps to solve a proportion:

1. Cross multiply
2. Distribute if there are parentheses
3. Get your variables on the same side of the equal sign
4. Solve

Solve each proportion. Show your steps.

1.

2.

3.

4.

Part 2: Absolute Value Equations

Remember: Absolute value is the distance from zero, so an absolute value equation usually has two answers: a positive and a negative.

Example: Solve $|x| + 1 = 6$.

Step 1: Get the absolute value by itself. Subtract 1 from both sides: $|x| = 5$

Step 2: Write both answers: **x = 5 or x = −5**

Solve each equation. Show your steps.

1. $|x| = 7$ x = _____ or x = _____

2. $|x| + 3 = 8$ x = _____ or x = _____

3. $|x| - 4 = 2$ x = _____ or x = _____

4. $2|x| = 8$ x = _____ or x = _____

Answer Key (Teacher Copy)

Part 1: 1) $x = 3$ 2) $x = 4$ 3) $x = 2$ 4) $x = 2$

Part 2: 1) $x = 7$ or -7 2) $x = 5$ or -5 3) $x = 6$ or -6 4) $x = 4$ or -4